ROGER LUO

Singapore (Student's Pass) | +65 8084 8140 | 2979570847@qq.com | github.com/LUOaini1213 | luoaini1213.github.io

SUMMARY

NUS MSc student (graduating Jan 2027) targeting Planning Engineer roles in semiconductor manufacturing. Trained in transport flow and capacity analysis; I build capacity, scheduling and inventory models in Python and SQL.

EDUCATION

National University of Singapore Aug 2025 – Jan 2027 (expected)

MSc in Civil Engineering (Transport and Urban Mobility)

Relevant modules: Advanced Construction Planning and Control, Lean Construction, Transportation Data Analytics and Modelling

Chang'an University Sep 2021 – Jun 2025

Bachelor of Engineering in Transportation (Excellence Engineer Program) | Operations Research, Systems Engineering

EXPERIENCE

China State Construction Far East Facade Singapore | Jun 2026 – Present

Engineering Intern, Project Engineering – semiconductor fab (Micron) and LTA office facade packages

- Maintain the list, status and delivery plan for 15 facade material types; track milestones on a board that links supplier deliveries to site installation, so late items are flagged early.
- Link the China-based project team with the main contractor: align delivery and installation schedules, coordinate testing with four suppliers, and report deviations and resource gaps to stakeholders weekly and monthly.
- Automated supplier email tracking with Python scripts that parse shipment and test emails into batch and result records, and built a React/Node.js/SQLite goods-receipt web app to keep material records consistent.

Chery International – *Product Intern* China | May 2026 – Jun 2026

- Built the full-cycle Gantt chart for a KD localisation project, defining milestones and the critical path; aligned R&D, production and logistics schedules for sample and production parts.

SELECTED PROJECTS

Personal projects built with AI coding assistance; results are reproducible and test-checked in the linked repositories.

Fab Capacity Planning: Intel Mini-Fab Oct 2026

Public Intel Five-Machine Six-Step Mini-Fab case | fab-capacity-planner

- Built a static capacity model of a re-entrant fab that reproduces the published utilisation table exactly (diffusion 87.9%, implant 77.8%, litho 91.3%); adding litho setups pushes litho to 103.1% and caps output at 81.5 starts/week.
- Compared it with a G/G/m queueing model (cycle time, X-factor) and a SimPy simulation with batching, setups, PM, failures and operators (10 replications, 95% CIs); ranked debottlenecking options by throughput gain per dollar.
- Wrote an 8-quarter capital (tool purchase) and hiring MILP in PuLP/CBC against a demand forecast, re-planned for +20% demand and a product-mix shift. The simulation showed the Q4 plan saturating at 76.1 starts/week against 79 demanded because litho waited for an operator 31.5% of the time; hiring one quarter earlier raised capacity to 92.5.
- Added an LLM agent that answers capacity questions from the models' outputs; a validator rejects any answer containing a number it cannot trace to a computed value.

Inventory Policy and Warehouse Slotting Oct 2026

Synthetic data | global-logistics-ops

- Simulated base-stock, (s,S) and (r,Q) policies for ocean versus air supply; a slotting heuristic cut picker travel by 21.2% versus ABC slotting on held-out orders (20 seeds, 95% CI 20.6–21.8%).

Fab Process Control and Tool Hookup Planning Oct 2026

fab-process-control | tool-hookup-planner

- SPC (X-bar/R, EWMA, CUSUM, Cp/Cpk) and PCA fault detection on the public SECOM fab dataset.
- Tool hookup planning on a synthetic sub-fab BIM model: utility capacity, HiGHS MILP, 3-D routing, clash checks.

Other Projects and Awards 2026

- Production planning with downtime replanning, HiGHS + SimPy (manufacturing-planning-agent).
- AYE weaving-section capacity and SUMO control scenarios, NUS group project (ce5203-aye-weaving).
- Civil Buddy: 3rd place (SME category), NUS-ISS Show Me Your Agents Hackathon 2026, 114 teams (civil-buddy-sme).

SKILLS

Planning and industrial engineering: fab capacity and capital planning, cycle time (X-factor, G/G/m queueing), LP/MIP (HiGHS, PuLP), discrete-event simulation (SimPy), inventory policy, SPC.

Data and tools: Python (pandas, NumPy, SciPy), SQL (MySQL, Hive), statistical analysis, Excel, Git, React/Node.js; AI-assisted coding.

Languages: English (professional), Mandarin (native).